

Date: 07 July 2026	Team ID: SB-7429
Project Name: Smart Lender: Applicant Credibility Prediction for Loan Approval	Maximum Marks: 5 Marks

SOLUTION ARCHITECTURE TEMPLATE

Solution Architecture Flow

The Smart Lender architecture spans across three key layers:

[Presentation Layer - HTML/CSS/JS] - home.html form takes 11 inputs and submits POST request to /predict. - result.html displays APPROVED/REJECTED decision and details.
▼ (HTTP POST JSON Payload)
[Application Layer - Flask app.py] - Extracts form parameters. - Preprocesses and scales LoanAmount (divides by 1000).
▼ (Scaled Input Array)
[Logic Layer - Model & Scaler Binaries] - Applies scaler.pkl to standardize numerical fields. - Runs loan_model.pkl (XGBoost) inference, predicting label and probabilities.
▼ (Decision & Probability Output)
[Presentation Layer - Result Panel] - Renders result template showing approval status, logs, and a return button.

Component Details Table

Component Name	Role in System	Technologies Used
Frontend Templates	Renders input controls, validation messages, decision outcomes, and EDA plots.	HTML5, CSS (Style.css), JavaScript, FontAwesome
Web Controller	Exposes API endpoints, parses HTTP POST parameters, calls scaler/model files, and passes values to template engine.	Python, Flask, Jinja2
Classifier Model	Fitted estimator predicting binary target (1: Approved, 0: Rejected) and probability metrics.	XGBoost, Pickle binary
Standard Scaler	Pre-fitted weights standardizing numerical incomes and loan sizes at prediction time.	Scikit-Learn StandardScaler, Pickle